

SUNDRE, AB, Canada

WMO: 712480

Lat: 51.778N

Lon: 114.683W

Elev: 3656

StdP: 12.86

Time Zone: -7.00 (NAM)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-27.8	-20.0	-35.0	0.9	-27.7	-26.6	1.4	-19.8	23.1	39.2	19.6	32.7	1.5	260	0.663

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	25.9	82.6	59.9	79.0	59.0	75.7	57.7	63.4	76.3	61.4	74.3	59.5	72.1	8.6	120

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
58.9	85.1	68.1	56.4	77.6	65.8	54.4	72.2	64.1	30.6	76.2	29.1	74.4	27.7	72.1	69.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
(a)	(b)	(c)	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
18.5	15.8	13.4	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
			-37.1	89.0	7.1	3.7	-42.2	91.6	-46.4	93.8	-50.4	95.8	-55.5	98.5
			-37.0	66.6	6.8	2.1	-41.9	68.1	-45.8	69.4	-49.6	70.5	-54.6	72.0

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	36.4	14.5	16.4	24.9	36.7	46.5	53.7	59.4	57.3	48.9	37.9	24.8	14.9
	DBStd	19.26	15.36	13.90	13.88	8.64	7.19	5.15	5.14	5.53	7.19	8.30	12.81	14.34
	HDD50	5732	1099	940	778	401	155	20	2	8	104	379	758	1088
	HDD65	10442	1564	1360	1243	848	575	341	186	243	483	839	1207	1553
	CDD50	782	0	0	0	3	45	131	293	234	71	5	0	0
	CDD65	16	0	0	0	0	0	1	11	4	0	0	0	0
	CDH74	1073	0	0	0	1	55	89	433	367	124	4	0	0
	CDH80	230	0	0	0	0	6	15	100	85	24	0	0	0
Wind	WSAvg	4.6	3.8	3.9	4.8	5.7	5.8	5.6	5.0	4.4	4.2	4.4	3.9	3.7
Precipitation	PrecAvg	20.1	0.6	0.5	0.9	1.3	2.6	3.8	3.4	2.8	2.0	0.8	0.6	0.6
	PrecMax	30.0	1.1	1.3	1.9	4.7	6.0	9.1	7.7	5.3	4.7	1.6	1.2	1.6
	PrecMin	12.1	0.0	0.0	0.3	0.1	0.8	0.8	1.0	0.4	0.3	0.3	0.2	0.1
	PrecStd	4.4	0.3	0.3	0.4	1.0	1.2	2.4	1.7	1.3	1.2	0.4	0.3	0.5
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	51.2	53.0	59.7	71.3	80.0	82.1	86.9	87.1	83.3	72.8	60.1	52.1
		MCWB	38.3	39.2	43.0	49.1	53.4	58.9	62.1	59.9	57.3	50.9	44.2	39.4
	2%	DB	46.6	46.8	53.8	64.9	74.4	76.2	82.7	82.0	77.6	65.1	52.9	46.1
		MCWB	36.1	35.6	39.5	46.0	52.2	56.7	61.7	59.4	55.9	48.4	40.9	35.4
	5%	DB	42.3	42.3	49.2	59.8	69.8	72.5	78.9	78.1	72.7	59.6	47.6	41.1
		MCWB	33.5	33.0	37.1	43.3	50.6	55.2	61.0	59.2	54.6	45.4	37.3	32.4
	10%	DB	37.2	37.7	44.9	54.4	65.2	69.0	75.2	74.2	67.5	54.6	42.7	36.1
		MCWB	30.6	30.2	35.1	40.6	48.6	53.6	59.5	58.2	52.5	42.7	34.0	29.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	39.4	39.6	43.8	49.9	56.1	62.0	67.3	64.8	60.1	52.9	45.1	39.8
		MCDB	49.6	51.5	57.5	67.2	72.5	75.8	78.9	78.0	77.2	70.1	57.6	51.0
	2%	WB	36.6	36.0	40.3	47.0	53.8	59.3	64.6	62.5	57.5	49.3	41.4	35.8
		MCDB	45.8	45.7	52.1	63.0	70.3	71.7	76.8	76.0	73.6	63.2	51.9	44.9
	5%	WB	34.0	33.3	37.7	44.1	51.7	57.0	62.5	60.6	55.4	46.3	38.2	32.7
		MCDB	41.9	41.6	48.4	58.2	66.7	68.8	75.5	73.6	70.6	58.1	46.7	40.0
	10%	WB	30.8	30.6	35.2	41.4	49.6	55.0	60.5	58.9	53.1	43.5	34.8	29.7
		MCDB	37.3	37.4	44.2	53.0	63.1	66.3	72.7	71.8	66.8	53.2	41.7	35.9
Mean Daily Temperature Range	5% DB	MDBR	27.5	29.6	25.6	24.5	26.2	24.0	25.9	27.1	27.4	26.5	25.2	26.6
		MCDBR	31.1	35.1	32.6	36.2	37.7	33.5	34.5	37.4	40.3	36.2	30.9	30.1
		MCWBR	21.9	24.7	20.1	19.3	18.1	16.4	16.5	17.8	20.7	20.6	19.5	20.8
	5% WB	MCDBR	30.4	34.0	30.9	33.8	33.5	28.2	29.9	31.9	37.0	34.4	29.8	29.0
		MCWBR	21.9	24.2	19.3	18.6	16.9	14.7	15.9	16.4	19.6	20.2	19.1	20.5
Clear-Sky Solar Irradiance	taub	0.254	0.262	0.290	0.335	0.344	0.352	0.347	0.361	0.316	0.280	0.255	0.241	
	taud	2.410	2.468	2.450	2.337	2.375	2.380	2.408	2.376	2.466	2.562	2.500	2.390	
	Ebn at Noon	249	281	291	286	288	286	285	275	277	271	250	235	
	Edh at Noon	18	23	28	36	36	36	35	34	28	20	16	15	
All-Sky Solar Radiation	RadAvg	344	643	1059	1445	1697	1791	1956	1624	1148	707	379	268	
	RadStd	20	44	64	108	134	142	128	111	132	70	25	17	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (54 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page